How do RNNs encode mixtures?

One network yields distinct geometries

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esther-poniatowski.github.io/ccn2026

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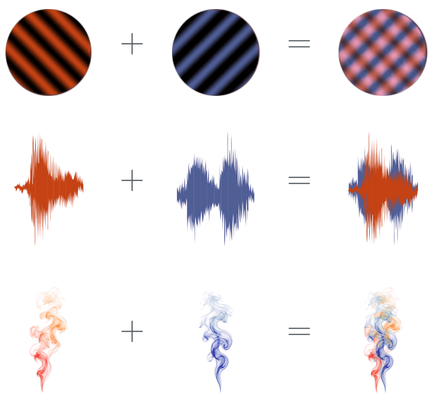
1 STARTING POINT

Natural inputs often mix simpler components.

Vision
overlapping gratings

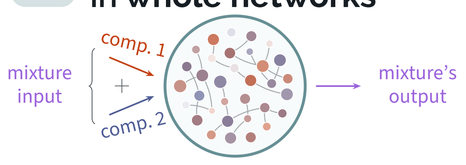
Audition
mixed voices

Olfaction
odor blends



GOAL

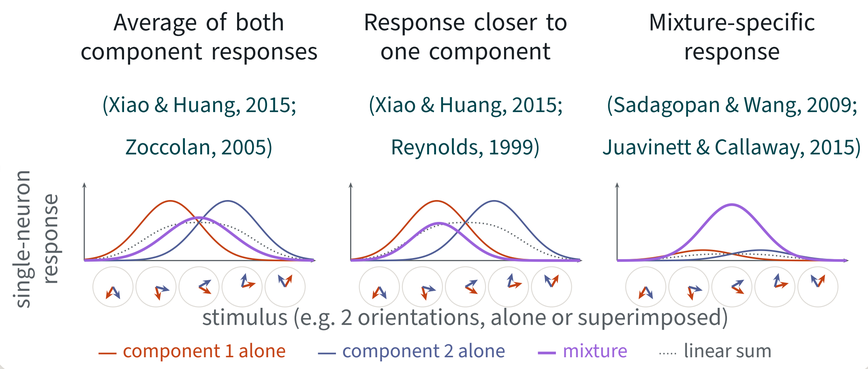
3 Understanding mixture coding in whole networks



TWO QUESTIONS

2 OBSERVATION

In single neurons, responses are heterogeneous.



4 QUESTION • Describing codes

Which distinct codes for mixtures are possible, *a priori*?

5 APPROACH • Describing codes

Geometry characterizes the code.

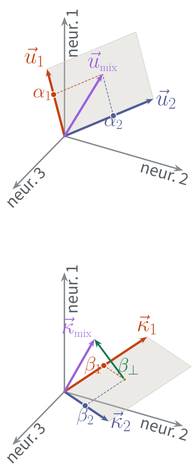
Across neurons, a mixture's output is one activity **vector**. Its geometry — its position *relative to* the component outputs — is what varies systematically.

Input vector

$\vec{u}_{\text{mix}} = \underbrace{\alpha_1\vec{u}_1 + \alpha_2\vec{u}_2}_{\text{linear sum of components}}$

Output vector

$\vec{\kappa}_{\text{mix}} = \underbrace{\beta_1\vec{\kappa}_1 + \beta_2\vec{\kappa}_2}_{\text{in the comp. plane}} + \underbrace{\beta_\perp\vec{\eta}_\perp}_{\text{orthogonal to plane}}$



LINEAR SUM
reference for characterizing geometry

By construction, the mixture input is the linear sum of its components.

The output may deviate from it by:
- a change of balance (β_1:β_2 against α_1:α_2)
- a part out of the plane ($\beta_\perp$)

7 QUESTION • Explaining codes

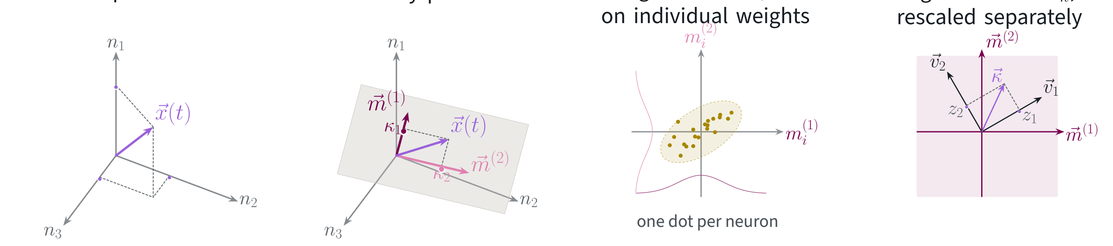
Which network mechanisms select the actual code?

8 APPROACH • Explaining codes

A reduced model links network structure to geometry. M 1

A few biologically plausible assumptions (Mastrogiuseppe & Ostojic, 2018; Beiran et al., 2021; Dubreuil et al., 2022) reduce an established neuron-level model to a few coordinates that capture output geometry.

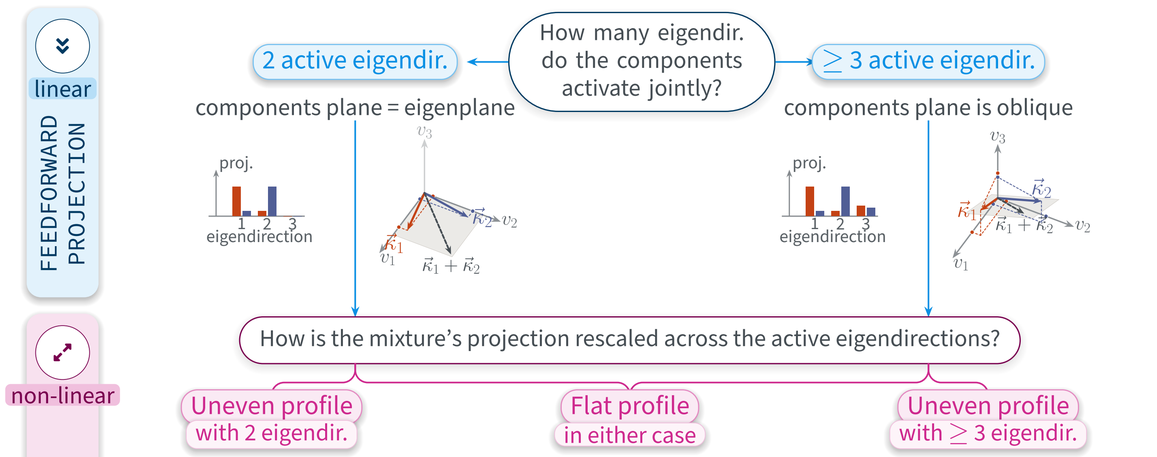
Standard rate model	Structured connectivity	Gaussian weight distribution	Change of basis
N neurons with external input and recurrent connectivity	sum of R rank-one matrices	in large networks ($N \to \infty$)	eigenvectors of the recurrent covariance
1	2	3	4
N coordinates, one per neuron	R coordinates, one per connectivity pattern $\vec{m}^{(r)}$	outputs depend on weight statistics, not on individual weights	R coordinates, one per eigendirection $\vec{v}_k$, rescaled separately



9 ANSWER • Explaining codes

Two successive transformations of the input control the geometry:

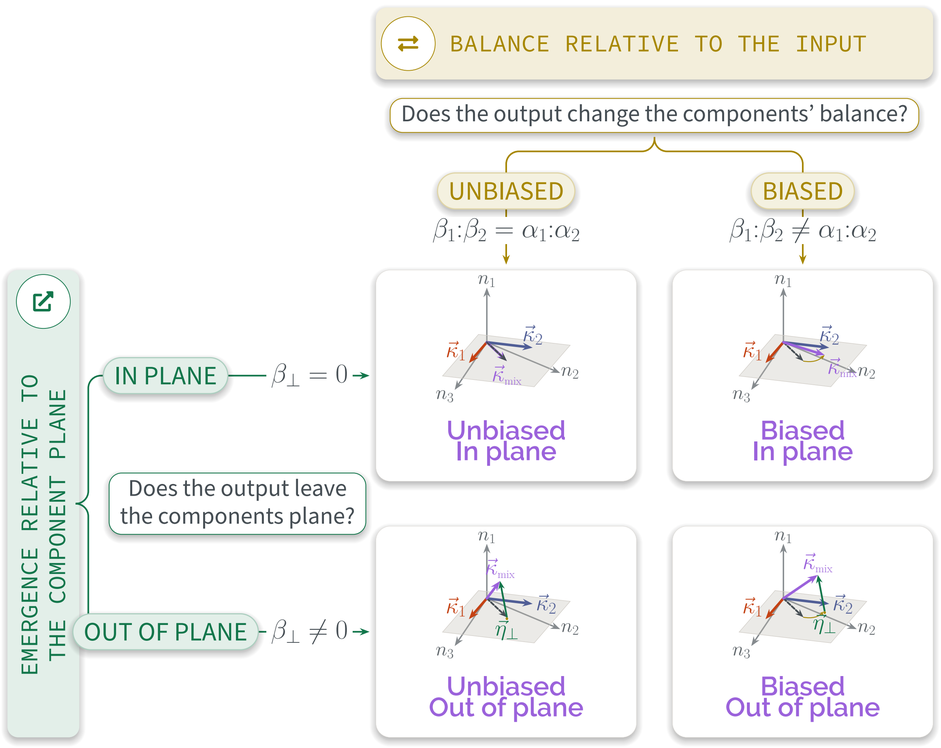
FEEDFORWARD PROJECTION & RECURRENT REWEIGHTING M 2 3



6 ANSWER • Describing codes

Mixture geometry varies along two independent properties:

BALANCE & EMERGENCE



10 INTERPRETATIONS

Each geometric property shapes what a readout can infer.

- A **biased** combination can implement attentional selection by **gain control**.
- An **emergent** pattern is **linearly separable** from its components — readable as a distinct identity.

11 PREDICTIONS

The model predicts testable signatures. M 4 5 6

- **Biased · Out of plane is the generic outcome**; the other geometries require special inputs or connectivity.
- **Emergence requires dissimilar components**, and grows as they separate.
- **Bias decreases** with stimulus intensity.

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